

Examiner
only

- 6. (a)** Plastics have replaced many traditional materials. The diagrams show plastics being used for three different types of food and drink packaging and containers.

Fruit packaging



Drinks containers



Take-away meal containers



Describe the advantages and disadvantages of using plastics in these examples.

[6 QER]



(b)

Microplastics in the Ocean

Plastics are used in many areas of modern life.

Figure 1 shows the amount of plastic production between 1950 and 2013.

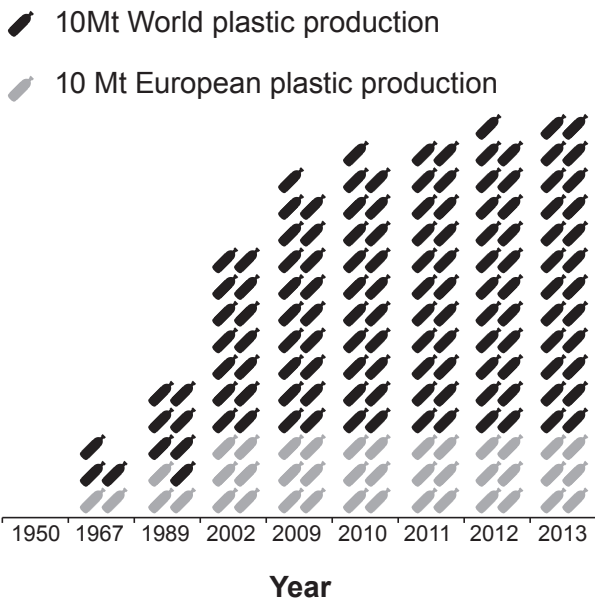
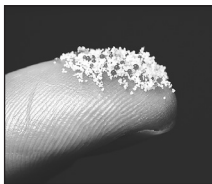


Figure 1

One of plastic's greatest properties, its durability, is also one of the main reasons that plastics present a threat to the marine environment.

The term 'microplastic' is used to describe plastic particles that are less than 5 millimetres in diameter, which includes particles as small as 10 nanometres. Microplastics can be found in some cosmetic products, toothpastes and soaps.



Microplastics are spread throughout the oceans and are found on shorelines from the Arctic to Antarctica. **Figure 2** shows the number of microplastic pieces found in sea ice at four Arctic sites during a survey in 2014.

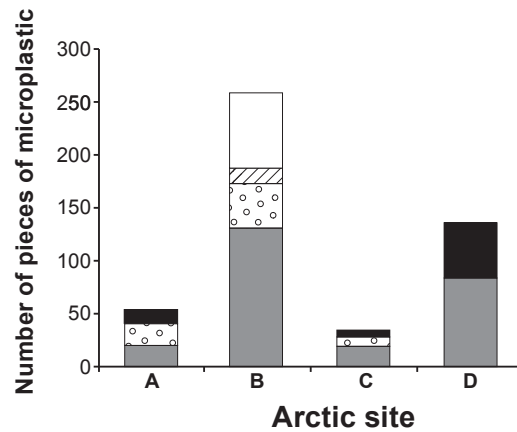


Figure 2

Key Plastic

-  nylon
-  polyester
-  rayon
-  polypropylene
-  others

Microplastics have been found inside the bodies of marine animals. Microplastics often contain chemicals that can absorb poisons such as pesticides from the surrounding seawater. There is strong evidence of transfer of poisons from eaten microplastics into animal tissues. Nano-size microplastics have been shown to cross cell membranes, under laboratory conditions, causing tissue damage.

Public awareness of the potential for microplastics to damage marine animals is low compared with that of the impact of plastic litter in our seas and oceans. Effective education of society is essential to raise awareness of the damaging effects of microplastics.



- (i) Put a tick (✓) in the box next to the size of microplastics. [1]

less than 10 mm

☐

between 5 mm and 10 nm

☐

greater than 5 mm and less than 10 nm

☐

between 5 mm and 10 mm

☐

- (ii) Put a tick (✓) in the box next to the statement that best describes the amount of plastic produced in Europe since 2002. [1]

plastic production has remained constant

☐

plastic production has increased

☐

plastic production has decreased

☐

- (iii) Name the type of plastic most often found in the Arctic microplastic survey. [1]

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- (iv) Put a tick (✓) in the box next to a hypothesis which needs further testing by scientists. [1]

the quantity of microplastics found in the Earth's oceans is increasing

☐

microplastics carry contaminants from sea water into animals

☐

microplastics cause tissue damage in marine animals

☐

microplastics are a greater problem near land than in deep water

☐

- (v) Suggest a method of educating people of the hazards of microplastics. [1]

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